

Chapter 51: Obesity — Revision Notes

Williams Obstetrics 26e, pp. 2350–2383 of the PDF. High-yield numbers are in **bold**. The three tables are transcribed from the PDF images (they are missing from the extracted chapter text). All 9 figures are included from the PDF.

Big picture: Obesity (now **>60%** of Parkland gravidas) raises almost every maternal, fetal and surgical risk in pregnancy — especially **hypertension/preeclampsia, GDM, stillbirth, cesarean delivery and wound infection**. Its offspring carry the risk into adult life. Bariatric surgery reverses much of this but brings its own nutritional and surgical hazards.

1. General Considerations

Definitions and prevalence

- **BMI (Quetelet index) = weight (kg) / height (m)².**
- US adult obesity prevalence (2017–2018): **42%**.

Category (NIH, 2000)	BMI (kg/m ²)
Normal	18.5–24.9
Overweight	25–29.9
Obese	≥30
Class 1 obesity	30–34.9
Class 2 obesity	35–39.9
Class 3 (morbid) obesity	≥40
Supermorbid obesity	≥50

- Obesity in girls and women rises with **age**, varies with ethnicity and worsens with **poverty**. A **genetic predisposition** exists.

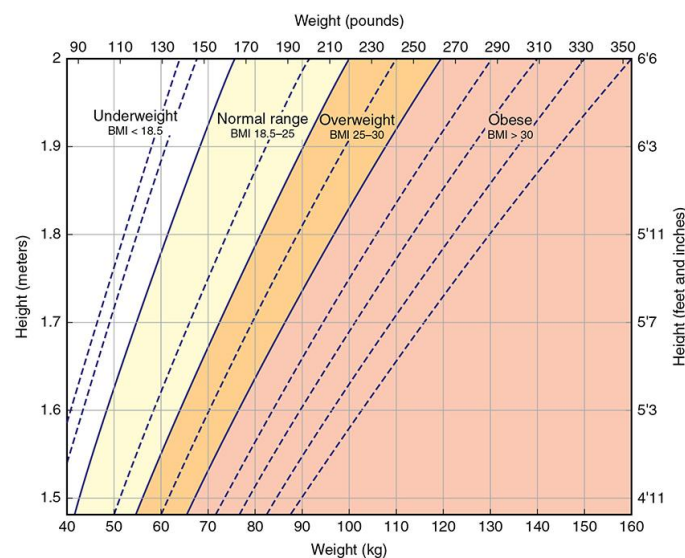


Figure 51-1. BMI chart: find where height and weight intersect to read the BMI category.

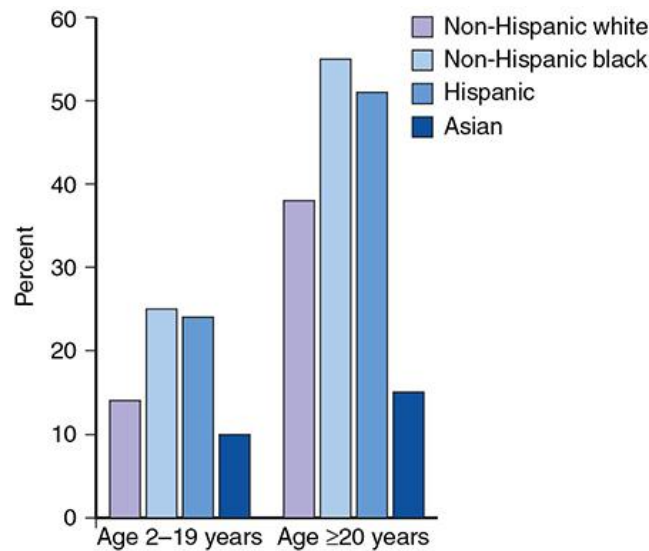


Figure 51-2. US obesity prevalence by race (2015–2016): highest in non-Hispanic black (~55%) and Hispanic (~51%) adults, lowest in Asian adults (~15%).

Adipose pathophysiology

- Fat is an **endocrine/paracrine organ**: adipokines, lipokines, exosomal microRNAs (adiponectin, leptin, TNF- α , IL-6, resistin, visfatin, apelin, VEGF, lipoprotein lipase, IGF).
- Adiponectin = the principal adipokine**:
 - ↑ insulin sensitivity, blocks hepatic glucose release, cardioprotective on lipids, anti-inflammatory.
 - Negatively regulated by fat mass**. Deficit → diabetes, hypertension, endothelial activation, CVD.
- Cytokines causing insulin resistance: leptin, resistin, TNF- α , IL-6** (all higher in pregnancy). Inflammatory adipokines may be the **primary stimulant of insulin resistance**.
- Perivascular fat also regulates vascular function; the placenta makes adipokines too.

Metabolic syndrome

- Obesity + inherited factors → **insulin resistance** → type 2 diabetes, dyslipidemia, hypertension → accelerated CVD.
- Diagnosis = any 3 of 5 criteria** (Table 51-1). **Waist circumference is the preferred screening measure**.
- Most type 2 diabetics meet the criteria. Obese hypertensive women typically have ↑ plasma insulin.
- US prevalence (NHANES, 2012): **34%** overall; **20%** at 18–29 y, **36%** at 30–49 y.

Table 51-1. Criteria for Diagnosis of the Metabolic Syndrome

Patients with three or more of the following:

Criterion	Threshold
Elevated waist circumference ^a	Country- and population-specific
Elevated triglycerides ^b	≥150 mg/dL
Reduced HDL cholesterol ^b	<40 mg/dL in males; <50 mg/dL in females
Elevated blood pressure ^b	Systolic ≥130 and/or diastolic ≥85 mm Hg
Elevated fasting glucose ^b	≥100 mg/dL

^aAccording to country- and population-specific thresholds. ^bThose with normal values while taking medications are considered to meet these criteria. From the National Heart, Lung, and Blood Institute, 2019.

Memory hook: "1-5-0, 1-3-0/8-5, 1-0-0, 4-0/5-0" — TG 150, BP 130/85, glucose 100, HDL 40 (men)/50 (women), plus waist.

Nonalcoholic fatty liver disease (NAFLD)

- Visceral adiposity correlates with hepatic fat → **hepatic steatosis (NAFLD)** → **NASH** → cirrhosis → hepatocellular carcinoma.
- **The major cause of cirrhosis and liver cancer**; US cost >\$100 billion/yr. Strongly linked with CVD (Ch 58).

Obesity-associated morbidity

- Glucose intolerance, hypertension, dyslipidemia, metabolic syndrome → MI, **atrial fibrillation**, heart failure, stroke.
- Adiposity accounts for **up to 75% of primary hypertension**.
- Insulin resistance → hippocampal changes, ↓ executive function and memory. Cytokine-induced **osteoarthritis**.
- ↑ All-cause early mortality; CVD and cancer mortality rise **proportionally with BMI**. An "**obesity paradox**" (survival advantage in some groups) is hypothesized, but weight normalization clearly helps.

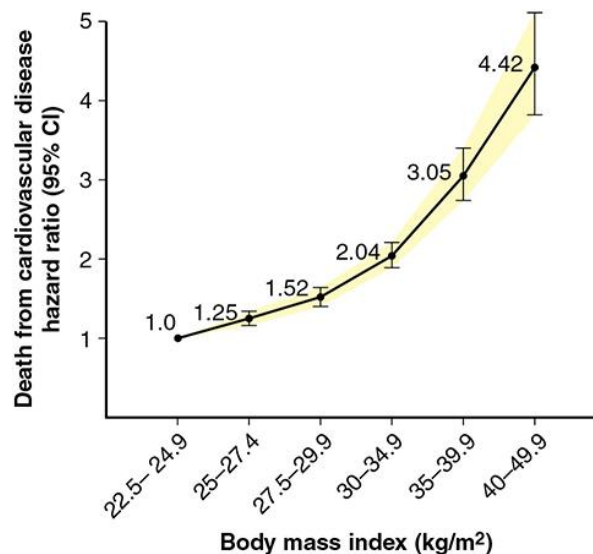


Figure 51-3. Cardiovascular death hazard ratio vs BMI (1.46 million adults): 1.0 at 22.5–24.9 → 2.04 at 30–34.9 → 4.42 at 40–49.9.

Obesity treatment

- **Weight loss is tremendously difficult**, and maintenance equally so. Ob-gyns should help obese women lose weight.
- Options: behavioral, pharmacological, surgical, or combinations. Diet + exercise ↓ weight and metabolic syndrome.
- Long-term failure is appreciable: **50% failure** in diabetics after bariatric surgery.

2. Pregnancy and Obesity

- Reproductive disadvantages: **infertility**, **early and recurrent pregnancy loss**, **preterm delivery**, and obstetric, medical and surgical complications.
- Also: ↑ **oral contraceptive failure** and associated thromboembolism; ↑ complications of second-trimester surgical abortion.
- Offspring (as infants and adults) have ↑ morbidity.
- **Obese women now make up >60%** of Parkland gravidas.

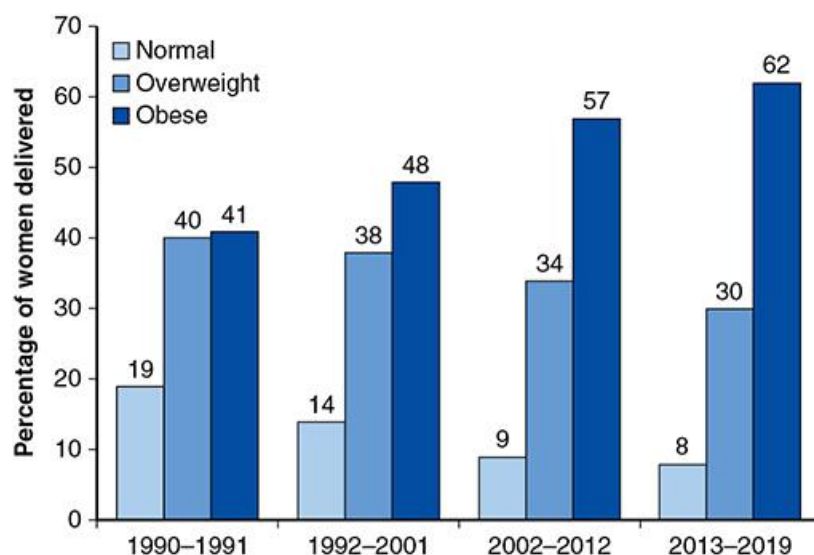


Figure 51-4. Parkland obesity at first prenatal visit: 41% (1990-91) → 62% (2013-19); normal-BMI women fell from 19% to 8%.

Maternal morbidity

- **Overweight women** already have significantly more complications than normal-BMI women (Table 51-2), though less than obese women.
- Obesity thresholds in studies range from >30 to >50 kg/m².
- **Maternal mortality nearly 4× higher** in obese women (Michigan).
- **Supermorbid obesity:** very high rates of preeclampsia, macrosomia, cesarean, meconium aspiration, ventilator support and neonatal death.

Table 51-2. Adverse Pregnancy Effects in Overweight and Obese Women

Complication	Normal BMI 18.5–24.9 (n = 621,048), %	Overweight BMI 25–29.9 (n = 228,945), % (OR ^a , 95% CI)	Obese BMI >30 (n = 78,043), % (OR ^a , 95% CI)
Gestational diabetes	2.3	4.3 (1.91, 1.86–1.96)	8.6 (4.04, 3.94–4.15)
Preeclampsia	2.7	4.3 (1.60, 1.56–1.64)	8.1 (3.17, 3.08–3.25)
Preterm birth	3.8	4.1 (1.09, 1.05–1.13)	4.8 (1.28, 1.23–1.34)
Labor induction	20.9	23.8 (1.19, 1.17–1.21)	29.7 (1.60, 1.57–1.64)
Prelabor or elective cesarean delivery	6.6	8.3 (1.28, 1.26–1.31)	11.5 (1.85, 1.81–1.89)
Cesarean delivery	25.2	31.5 (1.37, 1.34–1.39)	39.3 (1.92, 1.88–1.96)
Shoulder dystocia	2.0	2.4 (1.22, 1.17–1.28)	2.3 (1.14, 1.08–1.21)
Postpartum hemorrhage	6.7	8.4 (1.29, 1.26–1.31)	8.7 (1.34, 1.31–1.37)
Pelvic infection	0.6	0.7 (1.16, 1.06–1.26)	0.8 (1.28, 1.15–1.43)
Wound infection or complication	0.4	0.5 (1.42, 1.28–1.58)	1.0 (2.70, 2.42–3.01)
Large for gestational age	8.7	13.1 (1.57, 1.54–1.61)	16.3 (2.04, 1.99–2.10)
Macrosomia	2.0	3.6 (1.81, 1.74–1.88)	5.1 (2.60, 2.50–2.71)
Stillbirth	0.3	1.8 (5.89, 5.57–6.22)	0.5 (1.71, 1.56–1.87)

^aOdds ratios with 95% CI are significant when compared to normal BMI group. BMI = body mass index. Data from Kim, 2016; Lisonkova, 2017; Ovesen, 2011; Santos, 2019; Schummers, 2015. The stillbirth row is transcribed as printed; the overweight value (1.8%, OR 5.89) exceeding the obese value (0.5%) is inconsistent with the text, which says stillbirth rises with the degree of obesity.

Memory hook: The biggest odds ratios in obesity are **GDM (~4×)**, **preeclampsia (~3×)**, **wound complication (~2.7×)** and **macrosomia (~2.6×)**.

Hypertension and preeclampsia

- **Especially striking: hypertension and GDM rates.**
- Mechanism: insulin resistance → low-grade inflammation, endothelial activation, ↑ Na⁺ reabsorption → central to **preeclampsia**.
- Hypertension risk is **highest with BMI >50**.

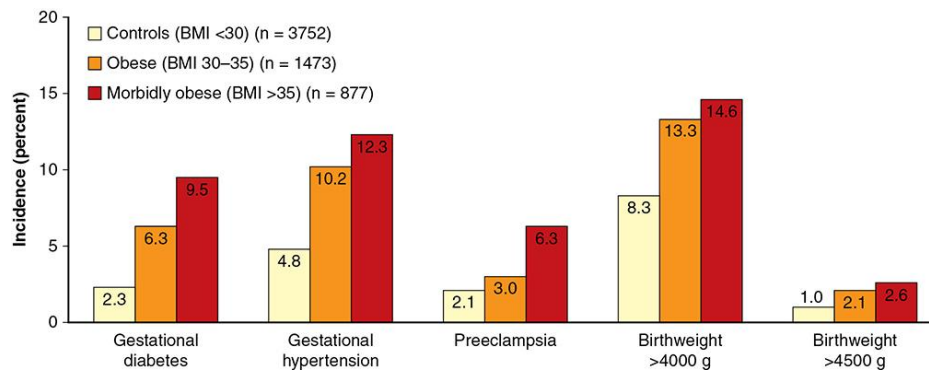


Figure 51-5. FASTER trial: GDM 2.3% → 6.3% → 9.5% and gestational hypertension 4.8% → 10.2% → 12.3% (controls → BMI 30-35 → BMI >35).

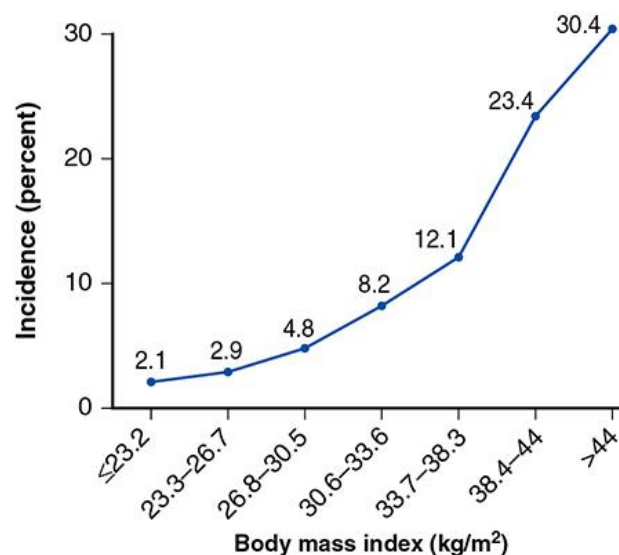


Figure 51-6. HAPO: preeclampsia rises steeply with BMI, from 2.1% (≤23.2) to 30.4% (>44).

Heart

- **Obesity + hypertension are common cofactors in peripartum heart failure.**
- Cardiac MR: **concentric remodeling** (abnormal) is greater in overweight/obese gravidas, but **regresses by 3 months postpartum**. Abnormal remodeling → ↑ hypertensive disorders.

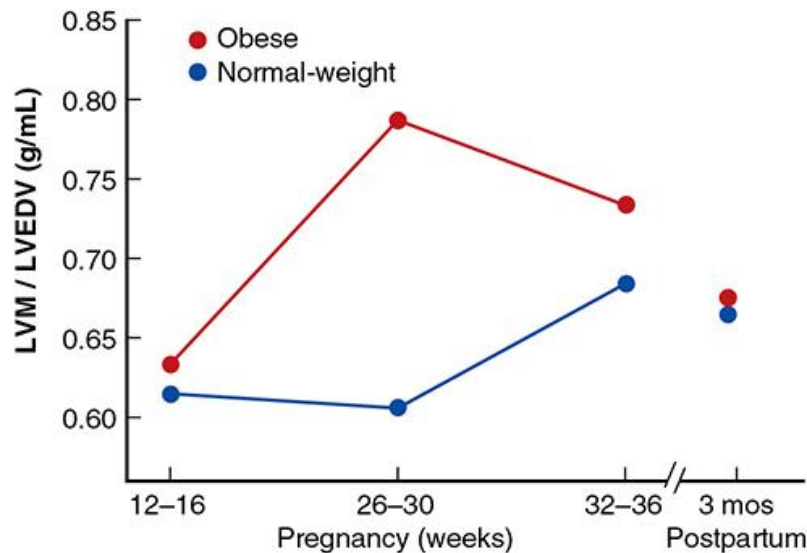


Figure 51-7. LVM/LVEDV ratio (concentric remodeling) peaks at 26–30 wk in obese women and returns to normal by 3 months postpartum.

Diabetes, liver and mental health

- **Obesity and GDM are inextricably linked** (Ch 60).
- **NAFLD in pregnancy** → ↑ preeclampsia, preterm birth, LBW, cesarean, GDM. **First-trimester sonographic NAFLD strongly predicts GDM.**
- **LDL-III predominance** (↑ in overweight/obese) = hallmark of ectopic liver fat (NAFLD).
- Steatohepatitis shows as ↑ transaminases; liver biopsy is rarely needed to exclude other causes.
- ↓ Quality of life; ↑ **depression** (during and after pregnancy) and **anxiety**.

Perinatal mortality

- **Stillbirth rises with degree of obesity.** Obesity is the **highest-ranking modifiable risk factor for stillbirth** (~100 studies).
- Supermorbid vs normal weight: stillbirth **5.7× at 39 wk and 13.6× at 41 wk. 25% of term stillbirths** involved obese women.
- Causes include chronic hypertension with superimposed preeclampsia; placental **decidual arteriopathy and infarcts**.
- ↑ Neonatal death. **Interpregnancy weight gain ↑ perinatal mortality**; weight loss between pregnancies (overweight women) ↓ it.

Perinatal morbidity

- Cofactors: **chronic hypertension and diabetes** → FGR and indicated preterm birth.
- PPROM in obese women → more neonatal respiratory complications.
- Pregestational diabetes ↑ birth defects; GDM → LGA/macrosomia.
- Even without diabetes or hypertension, neonatal morbidity is ↑. Preterm birth may relate to adipokine-driven chronic inflammation.
- **Prepregnancy BMI** (via inflammation and placental gene expression) has the **strongest influence on macrosomia** — more than gestational weight gain or diabetes.
- **Neural-tube defects: 1.2× (overweight), 1.7× (obese), 3.1× (severely obese).** BMI also correlates with congenital heart defects (possibly via diabetes).
- **Obesity degrades ultrasound accuracy** for anomaly detection.

Long-term offspring morbidity

- Obese mothers → **obese children** → **obese adults**.

- At ~9 years: childhood obesity, central obesity, ↑ systolic BP, insulin resistance, lipid abnormalities (all metabolic-syndrome elements).
- Adult offspring of overweight/obese mothers: ↑ **CVD and all-cause mortality** (n = 37,709). ↑ Glucose intolerance and metabolic syndrome.
- Excess maternal weight gain may predict adult offspring obesity.
- Mechanism unclear: **fetal programming**; epigenetics offers some support.

3. Antepartum Management

Maternal weight gain (IOM 2009; endorsed by ACOG)

Prepregnancy BMI	Recommended gain
Overweight	15–25 lb
Obese	11–20 lb

- **Weight loss during pregnancy is discouraged** (fetus, placenta, fluid and blood volume must be provided for).
- ⚠ These recommendations lack firm evidence. Conflicting data on inadequate gain:
 - No ↑ LBW/SGA with inadequate gain (47,494 obese women); weight loss did not impair fetal growth.
 - But gain below IOM → ~3× **SGA**; weight loss → ~2× **FGR**.
- **Excess gain**: ↑ maternal fat; ↑ preeclampsia, cesarean and fetal overgrowth in **overweight** (not obese) women. Overweight/obese women retain more weight postpartum.

Dietary intervention

- Exercise in overweight women ↓ **GDM** (one RCT of 300). Other trials: diet had no effect on gain.
- Cochrane (11,444 women): lifestyle interventions give only a **modest ↓ in weight gain**, with **no significant benefit** for overgrowth, cesarean or neonatal outcome.
- **Metformin in obese gravidas does not improve outcomes.**
- Why lifestyle fails: **introduced too late** — placental gene expression is programmed early.

Prenatal care

- Close monitoring detects early diabetes or hypertension. **Early GDM screening did not ↓ morbidity** vs standard screening.
- **Obstructive sleep apnea** is common.
- Standard anomaly screening is sufficient, remembering sonographic limitations. Ultrasound still estimates fetal weight accurately.
- **Serial sonography** is usually needed for growth: high specificity but **poor sensitivity for FGR** and low PPV for macrosomia.
- External FHR monitoring is harder.

4. Intrapartum Management

- ↑ **Postterm pregnancy** and labor abnormalities.
- Odds of **spontaneous labor at term ~half** those of normal-weight women (143,519 women).
- **BMI >30 → longer, slower early first stage.** Epidural does not lengthen labor in obese women.

Labor induction

- Obese women are **~2x as likely to be induced** (Table 51-2) and more likely to have a **failed induction** (risk rises with BMI). Induced labor lengthens with BMI.
- Conflicting data:
 - Nulliparas BMI >30 induced at 39 wk: cesarean **40% vs 26%** with expectant management.
 - Class III obesity: cesarean after failed induction **49%** overall (46% BMI 30–40, 63% BMI 40–50, 69% BMI >60).
 - But: elective induction at **37–39 wk** → **lower cesarean rate** (74,725 deliveries).
 - **Foley + misoprostol** → shorter labor and fewer cesareans than misoprostol alone in obese nulliparas.

Anesthesia risks

- **Difficult epidural/spinal placement** and failed/difficult intubation. **Anesthesia consult** for supermorbid obesity, during prenatal care or on arrival to labor unit (ACOG).
- Longer neuraxial procedure times and more failed attempts.
- Single-shot spinal and CSE are equally quick and effective in morbid obesity.
- Neuraxial hypotension → more **umbilical artery acidemia** (delayed delivery). **Cord pH <7.1: 3.5% (BMI <25) → 7.1% (BMI ≥40).**

Cesarean delivery

- Cesarean rates (226,958 women): **overweight 34%, class I 38%, class II 43%, class III 50%.**
- Obese women have **less composite morbidity when labor is attempted** than with cesarean.
- More **emergency cesareans** and longer decision-to-incision and delivery times.
- **Failed TOLAC is more frequent**; interpregnancy weight gain ↓ VBAC rates.

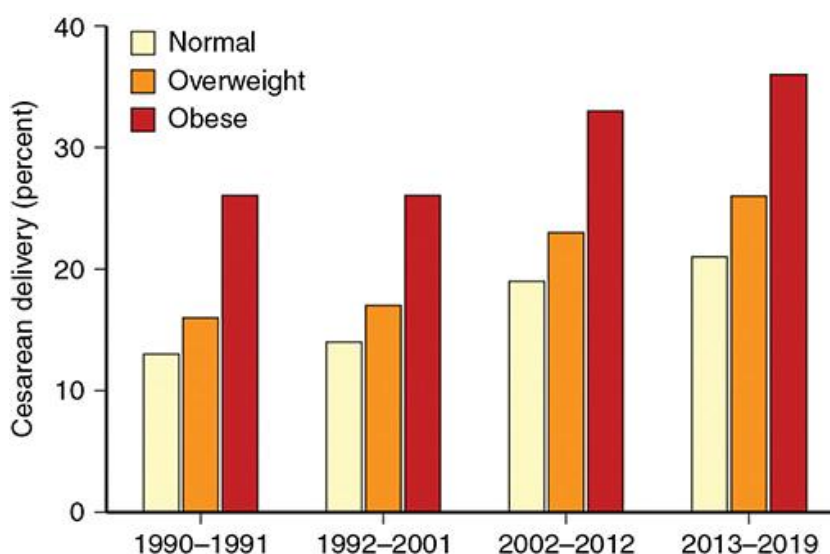


Figure 51-8. Parkland cesarean rates over 30 years: obese women consistently highest (~26% → ~36%).

Surgical concerns

Incision

- Parkland prefers a **vertical periumbilical incision** above the panniculus for direct access to the lower segment. Others use a low transverse incision ± rostral taping of the pannus. **No approach is proven superior.**

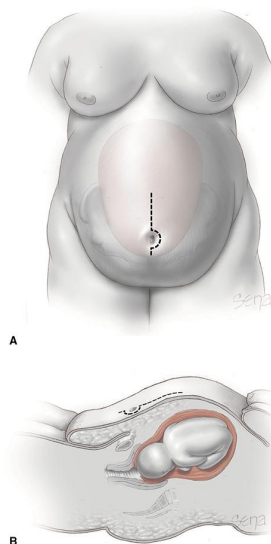


Figure 51-9. Incision in the obese woman: a periumbilical vertical skin incision above the panniculus (A, frontal; B, sagittal) gives access to the lower uterine segment.

Wound infection (directly related to BMI)

- Supermorbidity vs nonobese: **23% vs 7%** (3×). BMI >45: **14–19%** wound complications. Range 2 to >40% across studies. **Diabetes ↑ risk.**
- **Close subcutaneous tissue if ≥2 cm deep** → ↓ wound complications.
- **Staples = subcuticular closure** (identical results).

Antibiotic prophylaxis

- Tissue antibiotic levels fall with ↑ BMI. **Cefazolin 3 g** → higher tissue levels than 2 g, but **no fewer SSIs** (335 women, median 310 lb).

Source	Cefazolin dose
ACOG (2020)	2 g or 3 g for weight ≥80 kg
CDC	2 g for ≥80 kg; 3 g for ≥120 kg

- 2 g gives sufficient tissue levels for cesareans lasting **1.5 h**; consider **redosing** for longer surgery.
- **48 h oral cephalexin + metronidazole** postcesarean (added to routine prophylaxis): SSI **13.4% → 6.4%**.
- **Prophylactic negative-pressure wound therapy does not help**: 17% vs 19% (RCT of 441), confirmed later.

Other

- Puerperal tubal sterilization after vaginal birth is **safe regardless of BMI**.
- **Postpartum VTE risk ↑** (despite faster leg venous flow). Thromboprophylaxis is controversial (Ch 55):
 - **ACOG**: graduated compression stockings, hydration, **early mobilization** after cesarean.
 - **SMFM (2020)**: **enoxaparin** if risk factors **in addition to class III obesity**.
 - Parkland: **no routine thromboprophylaxis for obesity alone**.

5. Bariatric Surgery

- Two types: **restrictive** (↓ gastric volume) and **restrictive malabsorptive** (bypass).
- In nonpregnant patients: improve/resolve diabetes, hyperlipidemia, hypertension, OSA; ↓ MI and death.

Restrictive procedures

- **Laparoscopic adjustable silicone gastric banding (LASGB, LAP-BAND)**: band **2 cm below the GE junction** creates a small pouch; diameter set by a **saline reservoir**.
- After surgery: significantly ↓ **hypertension, GDM and preterm birth** in later pregnancies.

- **Band deflation:**
 - All 22 women fully deflated in the 1st trimester and gained **14.7 kg** on average.
 - Deflated vs inflated: gain **15.4 vs 7.6 kg**, birthweight **3712 vs 3380 g**, **2× macrosomia**.
- **Band slippage** (nausea/vomiting, advancing gestation, postpartum). One fetal death from **cerebral hemorrhage due to maternal vitamin K deficiency** after vomiting from band-slip gastric outlet obstruction.

Restrictive malabsorptive procedures

- **Laparoscopic Roux-en-Y gastric bypass = the most common procedure.** Remarkably ↓ hypertension, GDM and macrosomia (Table 51-3).
- **⚠ Upper abdominal pain** is frequent and may signal **internal herniation** (bowel through a mesenteric window defect):
 - Pain in **46%** of 139 pregnancies; **1/3 of these had internal herniation**.
 - Preterm birth **14/64 with pain vs 1/75 without**.
 - → **Intussusception, small bowel obstruction, maternal death**. Bowel obstruction is notoriously hard to diagnose in pregnancy.

Table 51-3. Pregnancy Outcomes Following Bariatric Surgery

Outcome ^a	Gastric Banding ^b	Gastric Bypass ^c
Hypertension	11%	4%
Gestational diabetes	7%	4%
Cesarean delivery	35%	33%
Mean birthweight	3206 g	3084 g
Low birthweight	7%	11%
Stillbirth	3/1000	3/1000

^aData not reported identically — frequencies are approximations. ^bData from Adams, 2015; Bar-Zohar, 2006; Carelli, 2011; Dixon, 2005; Ducarme, 2013; Facchiano, 2012; Lapolla, 2010; Pilone, 2014; Sheiner, 2009. ^cData from Adams, 2015; Ducarme, 2013; Facchiano, 2012; González, 2015; Kwong, 2018; Sheiner, 2009.

Pregnancy after bariatric surgery

- **Fertility improves** and obstetric complications decline vs morbidly obese controls.
- Almost **half are still obese** at first pregnancy after bypass (670 women), yet:
 - **LGA 22% → 8.6%**
 - **SGA 7.6% → 15.6%** (smaller fetuses from lower maternal glucose, not altered fetoplacental flow)
- ↓ Diabetes, preeclampsia and **congenital anomalies**.
- **ACOG management:**
 - Assess **vitamin and nutritional sufficiency**; supplement **B12, vitamin D, folic acid, calcium** as indicated. **Vitamin A** deficiency also possible.
 - **Gastric band:** follow with the bariatric team (band adjustments may be needed).
 - **Special vigilance for internal herniation with bowel obstruction.**

Pearl: Upper abdominal pain in a pregnant woman after Roux-en-Y bypass is internal hernia until proven otherwise — early surgical consultation.

Quick-Recall: Obesity vs Bariatric Surgery Effects

Outcome	Obesity (vs normal BMI)	After bariatric surgery (vs obese controls)
GDM	↑↑ (OR ~4)	↓
Preeclampsia / hypertension	↑↑ (OR ~3)	↓
Macrosomia / LGA	↑ (OR ~2.6 / 2.0)	↓ (LGA 22 → 8.6%)
SGA / FGR	↑ (via HTN, diabetes)	↑ (7.6 → 15.6%)
Stillbirth	↑ (5.7× at 39 wk, 13.6× at 41 wk, supermorbidity)	~3/1000
Congenital anomalies	↑ (NTD up to 3.1×)	↓
Cesarean	↑ (39% vs 25%)	~33–35%
Preterm birth	↑ (mostly indicated)	↓ (banding); ↑ with internal hernia pain
Specific hazards	Wound infection, failed induction, difficult anesthesia, VTE	Band slip, internal hernia, micronutrient deficiency (B12, D, folate, Ca, A, K)

Key Numbers to Memorize

- BMI: overweight **25–29.9**, obese **≥30**, class 3 **≥40**, supermorbidity **≥50**
- Metabolic syndrome: **3 of 5** — TG **≥150**, HDL **<40 M / <50 F**, BP **≥130/85**, FBG **≥100**, waist
- US obesity **42%**; metabolic syndrome **34%**; Parkland gravidas obese **>60%**
- Adiposity → **up to 75%** of primary hypertension
- Maternal mortality **~4×**; preeclampsia **30.4%** at BMI >44 (HAPO)
- Stillbirth **5.7×** (**39 wk**) / **13.6×** (**41 wk**) with supermorbidity obesity; **25%** of term stillbirths in obese women
- NTD **1.2 / 1.7 / 3.1×** (overweight / obese / severely obese)
- IOM gain: overweight **15–25 lb**, obese **11–20 lb**
- Cesarean: overweight **34%**, class I **38%**, class II **43%**, class III **50%**
- Cord pH <7.1: **3.5%** → **7.1%** (BMI <25 → ≥40)
- Wound infection supermorbidity **23% vs 7%**; close subcutaneous tissue if **≥2 cm**
- Cefazolin **2 g ≥80 kg, 3 g ≥120 kg** (CDC); redose if surgery **>1.5 h**
- Oral cephalixin + metronidazole **48 h** → SSI **13.4%** → **6.4%**; NPWT **17% vs 19%** (no benefit)
- LAP-BAND **2 cm** below GE junction; Roux-en-Y pain **46%**, $\frac{1}{3}$ internal hernia